

Cisco Provider Edge Router (PE) SOP for Alarm Code ALARM_2110

Node: Cisco-PE-Kolkata-WB-124 | **Vendor:** Cisco | **Severity:** Warning | **Alarm Code:** ALARM_2110

Description

This Standard Operating Procedure (SOP) outlines the steps to diagnose and resolve a Warning-level alarm on the Cisco Provider Edge Router (PE) device, Node ID: Cisco-PE-Kolkata-WB-124, with Alarm Code ALARM_2110.

Potential Root Causes

Cause	Probability
High CPU utilization on the router due to excessive routing protocol updates or other CPU-intensive processes.	High
Memory leak on the router, causing memory utilization to exceed threshold limits.	Medium
Routing protocol misconfiguration or issues with routing protocol adjacency.	Low

Diagnostic Commands

```
show processes cpu sorted | exclude kmem | exclude idle
show memory | include Total
show ip bgp summary
```

Resolution Steps

- Step 1: {'Step': '1. Check CPU utilization on the router and identify the process responsible for high CPU usage.', 'Action': "Execute the command 'show processes cpu sorted | exclude kmem | exclude idle' to determine the process causing high CPU utilization."}
- Step 2: {'Step': '2. Check memory utilization on the router and identify any memory leaks.', 'Action': "Execute the command 'show memory | include Total' to determine memory utilization."}
- Step 3: {'Step': '3. Verify routing protocol adjacency and configuration.', 'Action': "Execute the command 'show ip bgp summary' to verify routing protocol adjacency and configuration."}
- Step 4: {'Step': '4. Adjust CPU-intensive process or optimize routing protocol configuration if necessary.', 'Action': 'Adjust the CPU-intensive process or optimize routing protocol configuration as required to resolve the issue.'}
- Step 5: {'Step': '5. Verify memory utilization and CPU usage after resolution.', 'Action': "Execute the commands 'show memory | include Total' and 'show processes cpu sorted | exclude kmem | exclude idle' to verify memory utilization and CPU usage after resolution."}

Post-Check Verification

```
show memory | include Total  
show processes cpu sorted | exclude kmem | exclude idle  
show ip bgp summary
```